26900257 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Skill Enhancement Course (SEC) - MG5SECFSC300 - RESEARCH METHODOLOGY AND STATISTICS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. What is the primary purpose of applied research? [U] / [CO1]
2. Analyse the essential characteristics of a good sample with reference to representativeness, adequacy, and freedom from bias. [An] / [CO3]
3. How is class width determined? [U] / [CO4]
4. What is data collection? [K] / [CO2]
5. Differentiate between convenience sampling and snowball sampling based on participant recruitment. [An] / [CO3]
6. Mention two characteristics of conceptual research. [U] / [CO1]
7. Examine the major factors that influence the selection of a sampling design. [An] / [CO3]
8. What is grouped data? [U] / [CO4]

9. How is coefficient of variation used to compare variability? [U] / [CO5]

10. What is the importance of objectivity in research? [K] / [CO1]

11. Differentiate between population and unit of analysis with a suitable example. [An] / [CO3]

12. Explain a discrete variable with an example. [U] / [CO4]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **4** marks

13. Explain the major steps involved in conducting analytical research. [K] / [CO1]

14. Analyse the limitations of secondary data and explain the factors that should be examined before using it in research. [An] / [CO3]

15. Explain interval and ratio scales and distinguish between them. [An] / [CO4]

16. Explain the steps involved in formulating a research problem. [K] / [CO2]

17. Analyse the factors that influence a researcher's choice between primary and secondary data for a particular study. [An] / [CO3]

18. Explain how statistical methods can support objective forensic conclusions. [An] / [CO5]

19. Explain the importance of informed consent and confidentiality in research. [U] / [CO1]

[4x5 = 20]

Part C

Essay Type Questions

Answer any **1** questions

Each question carries **10** marks

20. Analyse the major types of probability sampling, namely simple random, systematic, and stratified sampling. Compare their selection procedures and suitability for different population structures. [An] / [CO3]

21. Analyse a given forensic dataset and select the most appropriate graphical method for its presentation, giving reasons for your choice. [A] / [CO4]

[10x1 = 10]